Combinational - Assignment

1 Implement the following functions using
i) 2:1 MUX and NOT gates
ii) 4:1 MUX

(a) $F(A,B,C) = \bar{A}C + B\bar{C}$

$= \bar{A}C(B+\bar{B}) + (A+\bar{A})B\bar{C}$

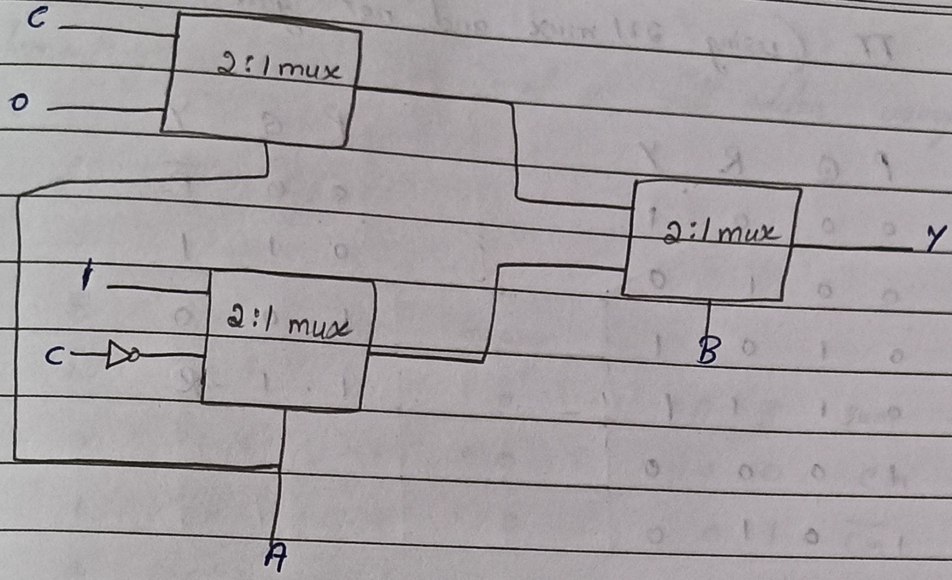
$= \bar{A}BC + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$

$= m(1, 2, 3, 6)$

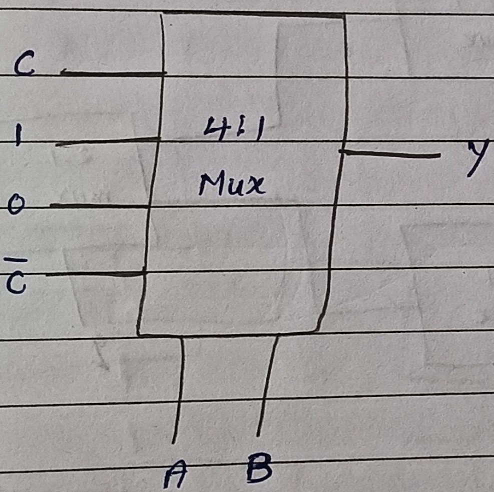
TT (using 2:1 mux and not gates)

A	B	C	Y (output)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

A	B	Y
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$



ii) using 4:1 mux (follow the above truth table)



b) $F(P,Q,R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

$= \bar{P}\bar{R}(Q+\bar{Q}) + (R+\bar{P})Q\bar{R} + \bar{P}Q(R+\bar{R})$

$= \bar{P}Q\bar{R} + \bar{P}\bar{Q}\bar{R} + PQ\bar{R} + \bar{P}Q\bar{R} + \bar{P}QR + \bar{P}Q\bar{R}$

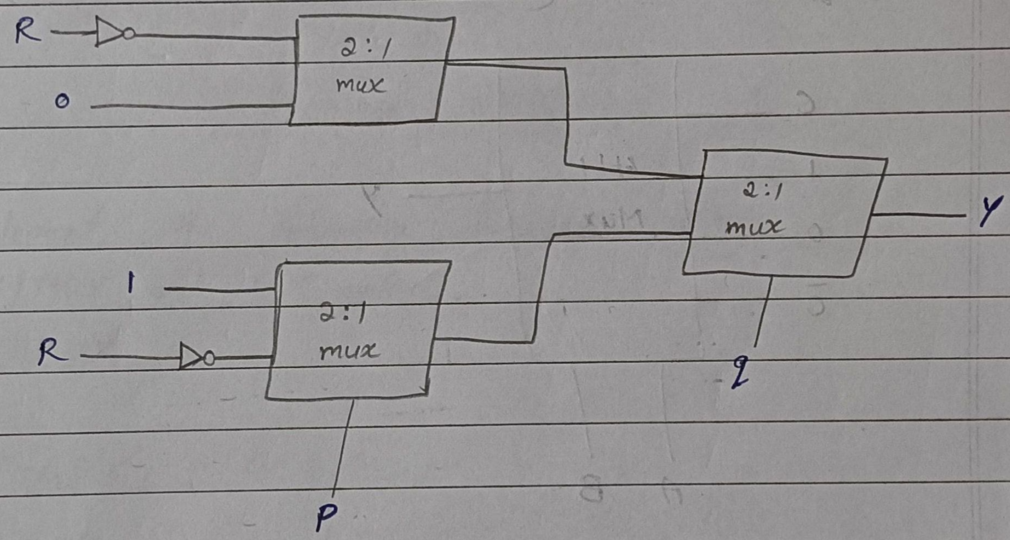
$= \bar{P}Q\bar{R} + \bar{P}\bar{Q}\bar{R} + PQ\bar{R} + \bar{P}QR$

$y = m(0, 2, 4, 6)$

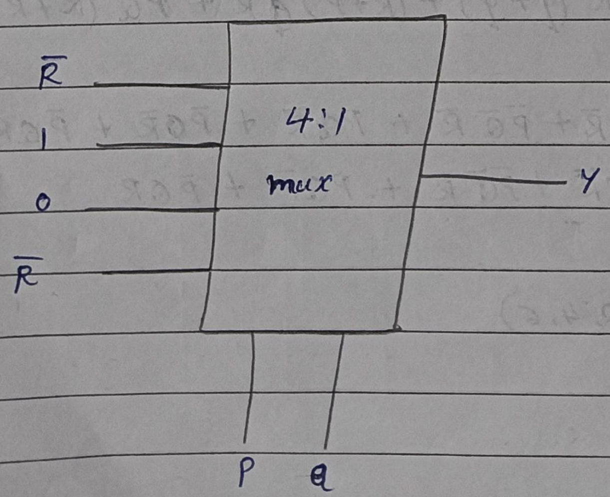
TT (using 2:1 mux and not gate)

P	Q	R	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

P	Q	Y
0	0	$\overline{R}$
0	1	1
1	0	0
1	1	$\overline{R}$



ii) using 4:1 mux

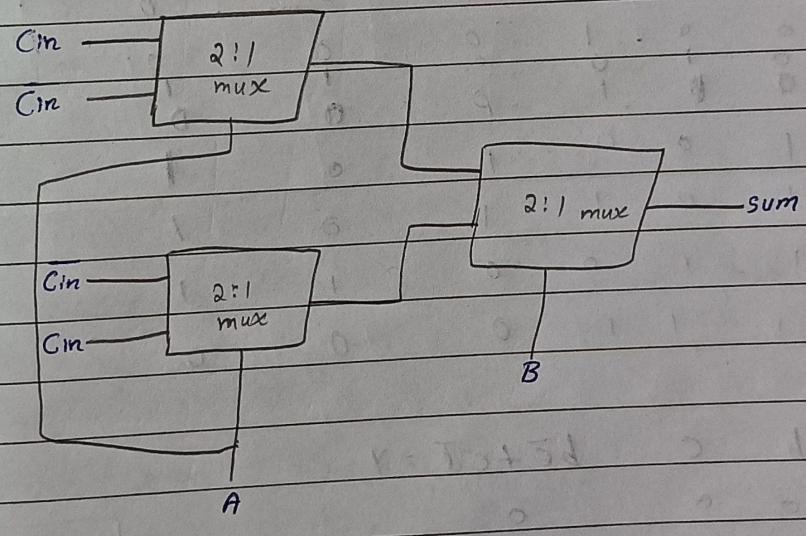


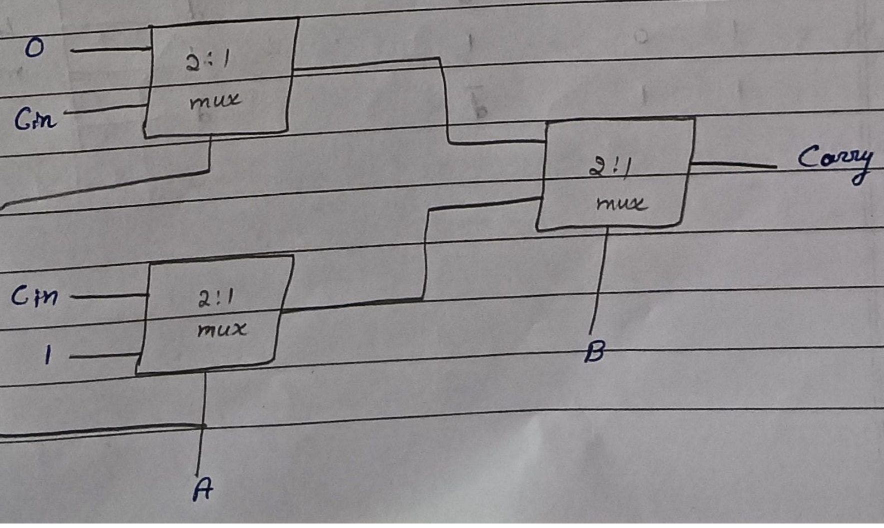
2 Implement the functionality of a full adder using 2:1 Muxes

TT of full adder

A	B	Cin	Sum	carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

A	B	sum	carry
0	0	Cin	0
0	1	$\overline{Cin}$	Cin
1	0	$\overline{Cin}$	Cin
1	1	Cin	1





3 Simplify $f(a,b,c,d) = \sum m(2,4,5,6,10) + D(12,13,14,15)$ and implement the SOP expression using 4:1 mux.

ab \ cd	00	01	11	10
00	0_0	0_1	0_3	1_2
01	1_4	1_5	0_7	1_6
11	X_{12}	X_{13}	X_{15}	X_{14}
10	0_8	0_9	0_{11}	1_{10}

$y = b\bar{c} + c\bar{d}$

Let b and c be select lines

b	c	d	$b\bar{c}$	$c\bar{d}$	$b\bar{c}+c\bar{d}$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

b	c	$b\bar{c}+c\bar{d} = Y$
0	0	0
0	1	$\bar{d}$
1	0	1
1	1	$\bar{d}$